SECTION 3
MAGNET RAMPDOWN (DECREASE TO ZERO)

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, *MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, *MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, *MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

3-1 INTRODUCTION

The magnet is initially ramped down until the helium level reaches 80%, then it is parked and refilled with liquid helium before finally being ramped down to zero field. Make sure the magnet's helium level is 100% before beginning this procedure. Also make sure to plan for the helium fill while the magnet is parked. The parked fill will use approximately 500 liters of liquid helium and take about one hour. A rampdown flowchart is shown in Illustration 3-1.

The 750 Amp Ramp Cable Kit (2135435) and Shim Cable Kit (2135558) are required to ramp and shim the magnet.

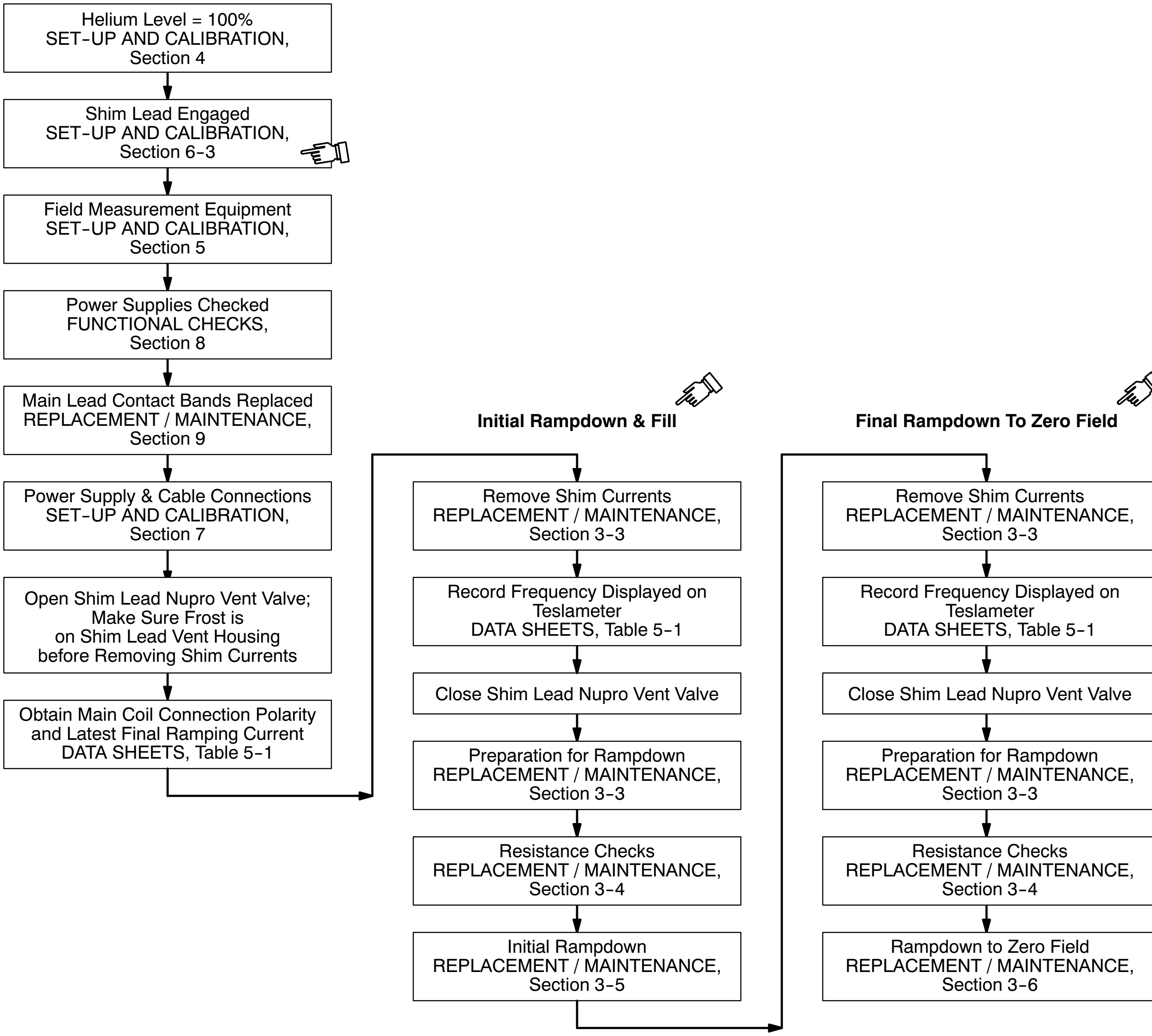
3-1 INTRODUCTION (continued)

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING DOWN THE MAGNET:

- **MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM.**
- **MAKE SURE THE MAGNET IS 100% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.**
- **POWER SUPPLIES MUST BE LOCATED OUTSIDE THE MAGNET ROOM TO RAMP DOWN THE MAGNET.**
- **REMOVE ALL FERROMAGNETIC OBJECTS, TOOLS OR EQUIPMENT FROM THE MAGNET ROOM. THEY CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**
- **A SUPERCONDUCTING MAGNET IS AN ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY DURING A QUENCH AND SENDING 100+ VOLTS ACROSS THE MAIN LEADS AND EXTENSIONS IF TOUCHED SIMULTANEOUSLY.**
- **MAKE SURE THAT INPUT POWER TO THE MAIN POWER SUPPLY IS DISCONNECTED WHEN CONNECTING THE MAIN LEADS AND THAT POSITIVE AND NEGATIVE POWER LUGS DO NOT CONNECT TO EACH OTHER.**
- **REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, *MR MAGNET - SAFETY REQUIREMENTS*, INCLUDED WITH THIS MANUAL.**

3-1 INTRODUCTION (continued)



FLOWCHART OF RAMPDOWN USING SHIM LEAD
ILLUSTRATION 3-1

3-2 TOOLS / EQUIPMENT

3-2-1 Safety Equipment

- 3.0T Warning Sign & Label Kit 2379494
- Portable Oxygen Monitor 2287000
- Cryogen Safety Wear Kit 46-271137G1
- Nonabsorbent protective clothing (long sleeve shirt and pants)
- Nonferrous Shoes

3-2-2 Other Tools / Equipment

- Main Coil Power Supply:
  - 750 Amp Model TCR 7.5T750 (46-260776G3) *or*
  - 1000 Amp Model ESS 7.5-1000-2-D-1236 (46-260776G4)
- Shimming Power Supply 46-260777G3
- 750 Amp Ramp Cart / Cable Kit 2135435
- Shim Case / Cable Kit 2135558
- Magnet Ramping Equipment Kit 46-260703G5 (for standard ceiling height sites), which includes:
  - Fixed Main Lead Extension 46-294204G1
  - Hold-Down Tool 2193922
  - Retractable Shim Lead Assembly 2285073
- 2.5M Ceiling Height Kit M1060SR (if required for low ceiling height sites), which includes:
  - Flexible Main Lead Extension 2287029
  - Hold-Down Tool, Short, 2276755
  - Retractable Shim Lead Assembly, Short, 2285073
- LCC300 Ramp Lead Clamping Fixture 5116737 (only for use when LHe topoff is planned immediately after parking the magnet in Section 3-5)
- Digital Voltmeter (DVM)
- MetroLab Teslameter Kit 46-251865G4 or equivalent magnetic field measuring device.
- No. 6 MetroLab Field Probe 2295525-5

3-2 TOOLS / EQUIPMENT (continued)
3-2-2 Other Tools / Equipment (continued)

- 50 Ft. (15.24 M) Auxiliary Rampdown Cable 2217315
- Nonmagnetic Tools Kit 46-320273G4

3-3 PREPARATION FOR RAMPDOWN

IMPORTANT !!!

Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.

1. "Top Fill" the magnet to = 100% helium level in conformance with SET-UP AND CALIBRATION, Section 4, Liquid Helium Fill, before continuing with this procedure.
2. Make sure the Shim Lead Assembly is "Engaged" in conformance with SET-UP AND CALIBRATION, Section 6-3, Shim Lead Engagement.

IMPORTANT !!!

If the Shim Lead cannot be engaged into the Shim Lead Connector (P100) in the magnet Vertical Penetration due to icing or damage to the P100's Sav-Con connector, use the Auxiliary Rampdown Cable (2217315) to ramp down the magnet. This cable connects the Main Coil Heater via the J1C connector on the magnet to J3 on the Main Power Supply. (See SCHEMATICS / INTERCONNECTS, Sections 2-1 and 3-1, respectively.)

Main coil voltage is not directly accessible for measurement with the Shim Lead disengaged. Use a digital voltmeter (DVM) with the Voltage Sense Leads connected to the Main Lead Extensions to read the approximate main coil voltage.

Use DVM readings during rampdown, especially when monitoring voltage in conformance with Table 3-1, Ramp Down Profile: Voltage Versus Current.

3. Set up the MetroLab Teslameter and Probe for single point field monitoring in conformance with SET-UP AND CALIBRATION, Section 5.
4. Perform the Ramp Power Supply checks in conformance with FUNCTIONAL CHECKS, Section 8, Main Power Supply Test. Verify that the Shim Supply Heater currents are set at 810 mA ± 10 mA.
5. Make sure the input power cables for the power supplies are disconnected.
6. Replace the Main Lead Contact Bands in conformance to REPLACEMENT / MAINTENANCE, Section 9, Main Lead Extension Contact Band Replacement.
7. Connect the Shim Power Supply to the magnet by making all cable connections in conformance with SET-UP AND CALIBRATION, Section 7-3, Superconducting Shim Coil Power Supply Connections to Magnet.

3-3 PREPARATION FOR RAMPDOWN (continued)

8. Open the Shim Lead Nupro Vent Valve and allow frost to appear on the Shim Lead Connector Housing before removing shim currents.
9. Connect the input power supply cable to the Shim Power Supply.

Note

This section allows one to remove all shim currents to find and document the actual main field to use as a target parking frequency when re-ramping the magnet. This will enable parking the magnet at a more accurate frequency and prevent retuning the rest of the system.

WARNING!

BEFORE CONTINUING THE RAMPDOWN PROCEDURE, MAKE SURE THE MAGNET CRYOGEN LEVEL IS > 95% TO PREVENT A POSSIBLE QUENCH.

MAKE SURE THE SHIM LEAD VENT CAP IS REMOVED AND FROSTING IS VISIBLE ON THE SHIM LEAD CONNECTOR HOUSING PRIOR TO TURNING ON THE SHIM POWER SUPPLY.

10. Set the Shim Power Supply's Main Power Switch to the "ON" position.
11. Set the Shim Power Supply to all last recorded Transverse 1 Currents from DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log. Make sure the current polarities are correct.
12. Set the Transverse 1 Heater Switch on the Shim Power Supply to "I" (on).
13. Verify that the heater current is 610mA ±10mA for T1 and T2 shims and 710mA ±15mA for axial shims. If it is not correct, adjust it with the adjustment screw located on the back of the Shim Power Supply. Allow a few minutes for the heater to drive the switches resistive.
14. Slowly adjust all Shim "Current" controls to zero.

Note

The field strength reading on the Teslameter should change as the Axial Shim Currents are adjusted down to zero. If no change in the Teslameter reading is noticed, the Axial Heater Switch may still be "persistent." If this occurs, wait an additional 2 more minutes before continuing to decrease currents.

15. Set the Heater Switch to "O" (off).
16. Repeat Steps 11 through 15 for the Transverse 2 and Axial Shim Coils.
17. Set the Shim Power Supply's Main Power Switch to "OFF" and disconnect all cables.

Note

The frequency displayed on the Teslameter after the shim currents are removed is the target frequency for when the magnet is reramped.

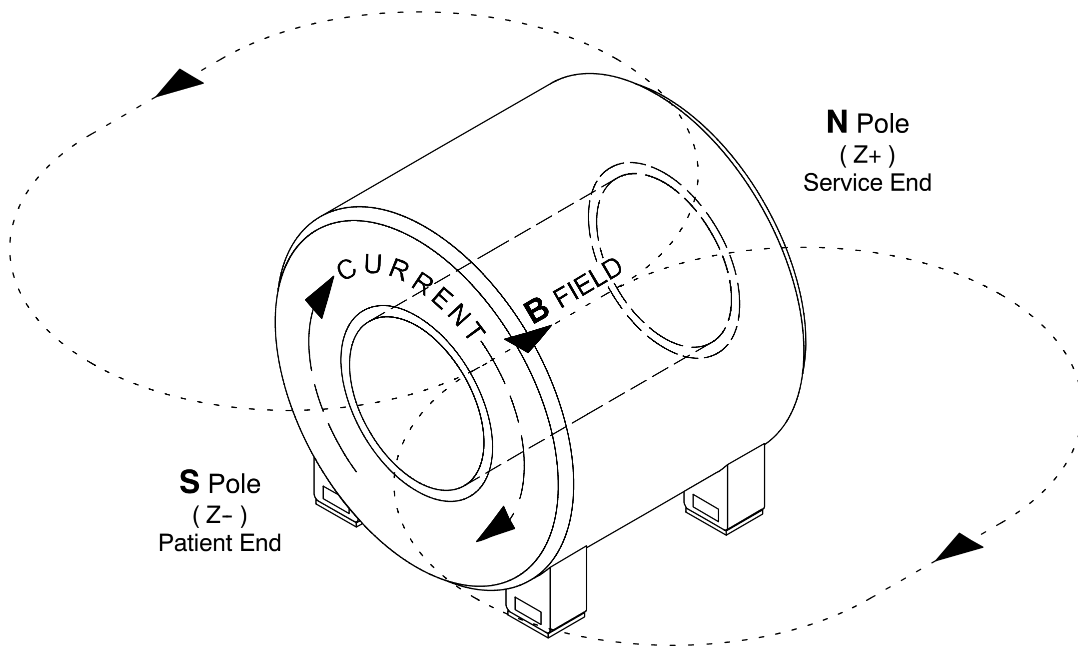
3-3 PREPARATION FOR RAMPDOWN (continued)

18. Record the frequency displayed on the Teslameter in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

19. Close the Shim Lead Nupro Vent Valve.

Note

All Main Coil driving voltages provided in this procedure will be equal in magnitude and opposite in polarity for magnets ramped with reversed polarity. Normal polarity is shown in Illustration 3-2.



NORMAL RAMPED MAGNET POLARITY
ILLUSTRATION 3-2

3-4 RESISTANCE CHECKS

1. Make sure the input power cable for the Main Power Supply is disconnected.

WARNING!

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW THEM TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBANT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

2. Connect the Main Power Supply and Main Lead Extensions to the magnet by making all cable connections in conformance with SET-UP AND CALIBRATION, Section 7-4, Main Coil Power Supply Connections to Magnet.
3. Set the Heater 1 Main and Heater 2 Shim Axial switches to the “O” (off) position.
4. Set the Current Adjust and Voltage Adjust controls to “O” (full CCW).
5. Connect the input power cable for the Main Power Supply.

WARNING!

MAKE SURE THE MAIN HEATER SWITCH IS OFF DURING RESISTANCE CHECKS. SETTING THE MAIN HEATER SWITCH TO “I” (ON) DURING THE RESISTANCE CHECKS WILL CAUSE A MAGNET QUENCH.

6. Set the Main Power and Power On switches to “ON.”
7. Set the Heater 2 Shim Axial Switch to “I” (on) and observe current rise in an analog ammeter (700 - 720 mA) to verify circuit continuity. Make sure the Main Heater Switch is off.

3-4 RESISTANCE CHECKS (continued)

8. Connect a digital voltmeter (DVM) to the end of the Voltage Sense Leads.
9. Set the Current Adjust Controls on the Magnet Power Supply to maximum (full CW).
10. Observe the Digital Current Meter and slowly turn the Voltage Control (CW) to set a 375A current through the Main Leads, Lead Extensions and persistent Main Switch.
11. Record the voltage reading on the DVM in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

WARNING!

A VOLTAGE READING > 100 MILLIVOLTS AT 375 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET, INCREASING BOIL-OFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.

12. If the DVM reads > 100mV, perform one or more of the following to reduce contact resistance:
    - Wait approximately 1 minute with the current running; the DVM readings may drop as the Main Lead Extensions cool.
    - If the reading still exceeds 100 mV:
      a. Turn the Voltage and Current Adjust Controls to zero (full CCW) and turn off the Magnet Power Supply input power.
      b. Check and tighten the bolts securing the Ramp Cables to the Power Supply and Main Leads Extensions.
      c. Lift and reseat the Main Leads.
    - Repeatedly failing the contact resistance check may indicate one of the following:
      a. The Main Lead Extension Contact Bands replaced as directed in Section 7-4, step 6, are not performing correctly and need replacing in conformance with REPLACEMENT / MAINTENANCE, Section 9.
      b. The Main Leads are damaged and need replacing.
    - Repeat Steps 1 through 11.
13. Continue to Step 14 after the DVM voltage is < 100mV.
14. Adjust the Voltage Adjust Control to pass 375A through the Main Leads, Lead Extensions and persistent Main Switch while observing the power supply's voltmeter. If the voltage exceeds 2.2V, discontinue the test and perform Step 12

3-4 RESISTANCE CHECKS (continued)

15. Set the power supply Voltmeter Select Switch to the “MAIN POWER SUPPLY” position. (This will display the output at the output lugs of the power supply monitored.) A voltage < 2.2V at 375A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 12 for adjusting contact resistance.

16. Turn the Current Adjust and Voltage Adjust controls off (full CCW) and continue with Section 3-5, Initial Rampdown & Fill.

3-5 INITIAL RAMPDOWN & FILL

WARNING!

- **THE MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND / OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS CHECKED FOR THE ABOVE AS REQUIRED DURING CALIBRATION. RIPPLE CAN ONLY BE ACCURATELY CHECKED AND ADJUSTED BY THE VENDOR.**
- **AXIAL SHIM SWITCH HEATERS <u>MUST</u> REMAIN ON DURING THE ENTIRE RAMP DOWN PROCESS TO PREVENT IRREPARABLE SHIM COIL DAMAGE AND A MAGNET QUENCH DURING RAMPING. THE POWER SUPPLY WILL NOT PASS CURRENT IN THE MAIN POWER LEAD CIRCUIT WITH THE AXIAL SHIM HEATERS OFF.**
- **MAKE SURE THE POLARITY RECORDED IN THE DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING CURRENT LOG, MATCHES THE CONFIGURATION OF THE RAMP POLARITY PLATE ON THE SHIM LEAD ASSEMBLY. THE “NORMAL” RAMP POLARITY IS “+” (POSITIVE) ON THE LEFT AND “-” (NEGATIVE) ON THE RIGHT AS VIEWED FROM THE COLDHEAD SIDE OF THE MAGNET. “REVERSED” RAMP POLARITY HAS “+” ON THE RIGHT AND “-” ON THE LEFT AS VIEWED FROM THE COLDHEAD SIDE OF THE MAGNET. IF THE PLATE AND DATA SHEET RECORD DO NOT MATCH, CALL THE NATIONAL SUPPORT CENTER TO AVOID A QUENCH.**
- **MAKE SURE THE CONNECTION POLARITY AND POWER SUPPLY CURRENT ARE THE SAME AS THE LAST RECORD IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING CURRENT LOG. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.**

Note
Normal polarity is shown in Illustration 3-2.

3-5 INITIAL RAMPDOWN & FILL (continued)

CAUTION

If a quench occurs during a change of the magnetic field, immediately turn the Main Power Supply Voltage Control and Current Control to zero, turn off the Power Supply and leave the room until the quench is over.

Note

Ice will build up around the Ramp Lead Hold-Down Tool flow holes during ramping down. Remove ice as needed to maintain a helium gas flow through the flow holes.

1. Retrieve the Main Coil connection polarity and latest final parking current from DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.
2. Make sure the Axial Shim Heater is on and the Main Leads are connected with the polarity indicated in Step 1.

Note

The Axial Shim Heater must remain on throughout the rampdown procedure.

3. Set the power supply Voltmeter Select Switch to “MAIN COIL” position.
4. Set the Current Adjust Controls on the Main Power Supply to10 amps more than the parking current retrieved in Step 1.
5. Set the Voltage Control to adjust the power supply output current to the parking current value obtained in Step 1 above.
6. Set the Heater 1 Main Switch to “I” (on).
7. Allow approximately 1 minute for the main switch to go normal.
8. When the main switch is normal, **slowly** decrease the Voltage Control (CCW) until negative 0.3 volts (-0.3 V) is observed across the power supply voltmeter (“MAIN COIL” position).
9. Continue to adjust power supply voltage using the Voltage Control during rampdown in conformance with the voltages and main coil current ranges shown in Table 3-1.

3-5 INITIAL RAMPDOWN & FILL (continued)

TABLE 3-1
RAMP DOWN PROFILE: VOLTAGE VERSUS CURRENT

Main Coil Voltage	Main Coil Current Range
-0.30 Volts	Start
-0.50 Volts	360 - 342 Amps
-1.00 Volts	342 - 314 Amps
-2.00 Volts	314 - 283 Amps
-4.00 Volts	283 - 248 Amps
-6.00 Volts*	248 - 0 Amps
Gradually turn the Voltage Control to zero (full CCW) then turn the Current Controls to zero (full CCW)	At 0 Amps
* The voltage will approach zero as the current approaches zero.	

10. Closely monitor the liquid helium level in the magnet during rampdown. When the level reaches 80%, immediately stabilize the field, park the magnet and disconnect / remove ramping connections and apparatus as follows:

Note
When the magnet is discharging (ramping down) a negative voltage is created by the decreasing current in the main coils (Ldi/dt). To stabilize the field increase the power supply voltage to compensate for the negative voltage. As the current change approaches zero the power supply voltage is reduced to maintain a stable current and field until it approaches < 0.1 volts to compensate for the I*R voltage drop in the circuit.

a. Slowly adjust the Voltage Adjust Control to stabilize the magnet field. The last two digits on the Teslameter should be the only digits changing.

b. Maintain the field at the stable setting for 5 minutes before proceeding to Step 10c.

Note
Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in the “Main Coil” position. When the switch goes ’persistent’ the voltage across the magnet terminals will stabilize at 0.00.

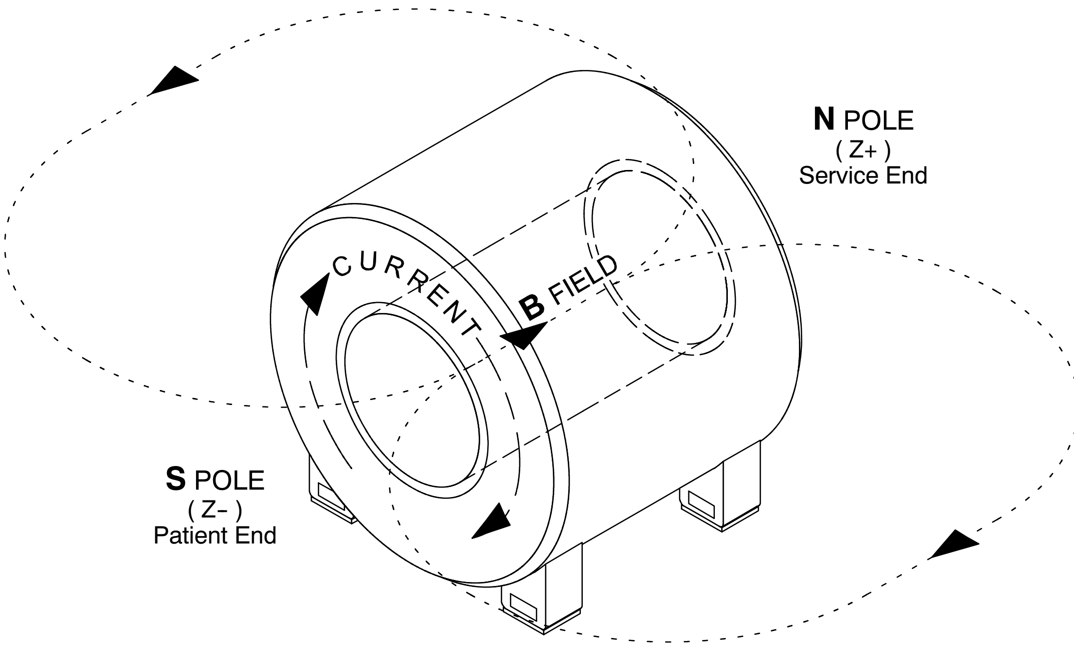
c. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go ’persistent.’

3-5 INITIAL RAMPDOWN & FILL (continued)

10. (continued)

WARNING!

MAKE SURE THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING CURRENT LOG. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH. (NORMAL POLARITY IS SHOWN IN ILLUSTRATION 3-3.)



NORMAL RAMPED MAGNET POLARITY
ILLUSTRATION 3-3

d. Record the current, frequency and lead extension voltage values at which the switch went 'persistent' in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

e. When the switch goes "persistent", **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

Note
Check that the Teslameter reading does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

3-5 INITIAL RAMPDOWN & FILL (continued)

10. (continued)

    f. Gradually turn the Current Adjust Control to zero (over a one minute period).

    g. Set the Heater 2 Shim Axial Switch to "O" (off).

    h. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

IMPORTANT !!!

LCC300 Ramp Lead Clamping Fixture 2343081 allows for immediate the helium fill while the Main Lead Extensions are in place. If this clamping fixture is being used, skip Steps 10i and 10j.

    i. Remove the Ramp Lead Hold-Down Tool (except if LCC300 Ramp Lead Clamping Fixture 5116737 is being used).

WARNING!

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBANT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

    j. Remove the Main Lead Extensions and cap the Ramp Lead Ports (except if LCC300 Ramp Lead Hold-Down Tool 2343081 is being used).

11. Fill the magnet to 100% in conformance with SET-UP AND CALIBRATION, Section 4, Liquid Helium Fill. The magnet will require approximately two 250 liter Dewars of liquid helium, and the fill will take approximately one hour.

3-5 INITIAL RAMPDOWN & FILL (continued)

12. Repeat Sections 3-3, Preparation for Rampdown, with the magnet at 100% helium level.
13. Repeat Section 3-4, Resistance Checks, with the magnet at 100% helium level.

3-6 RAMPDOWN TO ZERO FIELD

CAUTION

If a quench occurs during a change of the magnetic field, immediately turn the Main Power Supply Voltage Control and Current Control to zero, turn off the Power Supply and leave the room until the quench is over.

Note

Ice will build up around the Ramp Lead Hold-Down Tool flow holes during ramping down. Remove ice as needed to maintain a helium gas flow through the flow holes.

1. Retrieve the Main Coil connection polarity and latest final parking current from DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.
2. Make sure the Axial Shim Heater is on and the Main Leads are connected with the polarity indicated in Step 1.

Note

The Axial Shim Heater must remain on throughout the rampdown procedure.

3. Set the power supply Voltmeter Select Switch to "MAIN COIL" position.
4. Set the Current Adjust Controls on the Main Power Supply to10 amps more than the parking current retrieved in Step 1.
5. Set the Voltage Control to adjust the power supply output current to the parking current value obtained in Step 1 above.
6. Set the Heater 1 Main Switch to "I" (on).
7. Allow approximately 1 minute for the main switch to go normal.
8. When the main switch is normal, **slowly** decrease the Voltage Control (CCW) until negative 0.3 volts (-0.3 V) is observed across the power supply voltmeter ("MAIN COIL" position).
9. Continue to adjust power supply voltage using the Voltage Control during rampdown in conformance with the voltages and main coil current ranges shown in Table 3-1.
10. Observe the voltmeter reading with the power supply Voltmeter Select Switch in the "MAIN COIL" position. A zero reading on the power supply's digital voltage and current meters indicates that the magnet is fully discharged.

3-6 RAMPDOWN TO ZERO FIELD (continued)

11. When the magnet is fully discharged:

    a. Set the Heater 1 Main and the Heater 2 Shim Axial switches to “O” (off).

    b. Set the Main Power and Power On switches to “OFF.”

    c. Disconnect the input power cables from the Magnet Power Supply.

Note
The Pressure Controller will maintain helium vessel pressure.

WARNING!

A MAGNET AT ZERO FIELD CAN STILL CAUSE CRYOGEN BURNS. WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.

CAUTION

Replace the Ramp Port Caps immediately after removing the Main Lead Extensions to prevent ice-build up inside the Vertical Penetration.

Note
The steps below are performed <u>after</u> the magnet is ramped down and <u>after</u> AC power has been disconnected from the ramp power source.

12. Replace the flow hole screws in each Main lead Extension, **<u>one at a time</u>**.

    a. Install a screw into the flow hole of only one of the Main Lead Extensions. See Illustration 3-4.

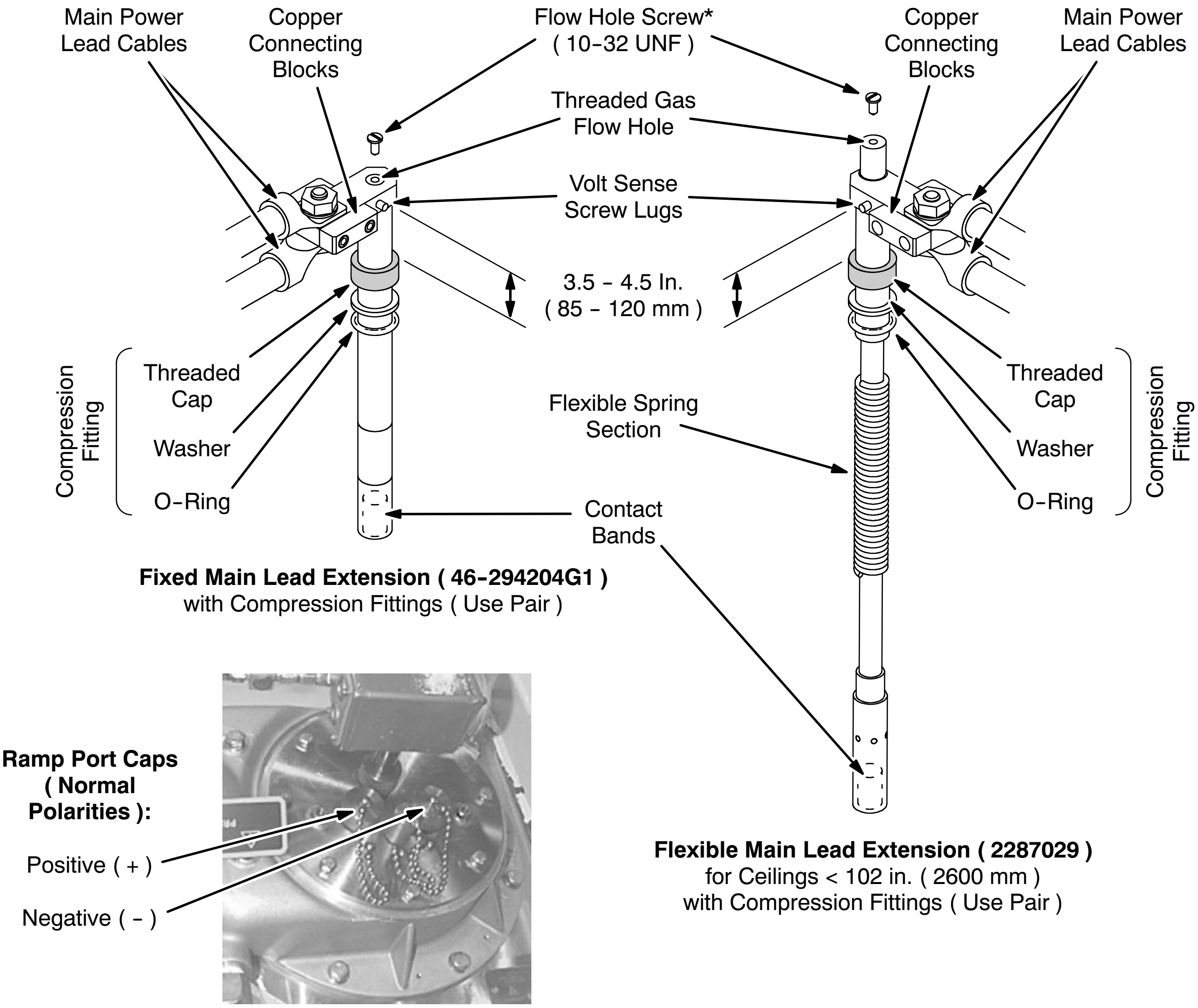
    b. Remove all ice around the Ramp Lead Port compression nut on the Main Lead Extension that is being removed (i.e. the Main Lead Extension that has the flow hole plugged in Step 12a). See Illustration 3-4.

    c. Unscrew the Ramp Lead Port compression fitting and remove the Main Lead Extension from the magnet.

    d. Immediately replace the cap onto the Ramp Lead Port.

    e. Repeat Steps 12a through 12d for the other Main Lead Extension.

3-6 RAMPDOWN TO ZERO FIELD (continued)



Fixed Main Lead Extension (46-294204G1)
with Compression Fittings (Use Pair)

Flexible Main Lead Extension (2287029)
for Ceilings < 102 in. (2600 mm)
with Compression Fittings (Use Pair)

MAIN LEAD EXTENSION AND RAMP PORT COMPRESSION FITTING
ILLUSTRATION 3-4